

Lab task#1

Task#1:

Transverse of Linear Array operations.

Input:

```
# include<stdio.h>
int main(){
    int arry[]={1,2,34,4};
    int i;
    for(i=0;i<4;i++)
    {
        printf("%d\n",arry[i]);
    }
    return 0;
}
```

Output:

```
PS C:\Users\zaem\.vscode> cd "c:\Users\zaem\.vscode\" ; if ($?) { gcc zaem.c -o zaem } ; if ($?) { .\zaem }
1      2      34      4
PS C:\Users\zaem\.vscode>
```

Insertion:

Input:

```
#include <stdio.h>

int main() {
    int arr[100] = {0};
    int i, pos, n = 10;

    for (i = 0; i < 10; i++) {
        arr[i] = i + 1;
    }

    printf("Original array: ");
    for (i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    pos = 5;
    n--;

    for (i = pos - 1; i < n; i++) {
        arr[i] = arr[i + 1];
    }

    printf("Array after deletion: ");
```

```

    for (i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    return 0;
}

```

Output:

```

PS C:\Users\zaem\.vscode> cd "c:\Users\zaem\.vscode\" ; if ($?) { gcc zaem.c -o zaem } ; if ($?) { .\zaem }
Original array: 1 2 3 4 5 6 7 8 9 10
Array after deletion: 1 2 3 4 6 7 8 9 10
PS C:\Users\zaem\.vscode>

```

Deleting:

Input:

```

#include <stdio.h>
int main() {
    int arr[100] = {0};
    int i, x, pos, n = 10;

    for (i = 0; i < 10; i++) {
        arr[i] = i + 1; }

    printf("Original array: ");
    for (i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
    pos = 5;
    n++;
    x=50;
    //insertion
    for (i = n-1; i>=pos; i--) {
        arr[i] = arr[i - 1];
    }
    arr[pos-1]=x;
    printf("Array after insertion: ");
    for (i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
    return 0;
}

```

Output:

```

PS C:\Users\zaem\.vscode> cd "c:\Users\zaem\.vscode\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Original array: 1 2 3 4 5 6 7 8 9 10
Array after insertion: 1 2 3 4 50 5 6 7 8 9 10
PS C:\Users\zaem\.vscode>

```